



MTH404

SOLVED QUESTIONS



Q.No.1: Define Apsidal Angle?

ANSWER: The angle between two consecutive apse lines is called an Apsidal angle.

Q.No.2: Derive the formula of maximum horizontal range.

SOLUTION: As we know that

$$\text{Horizontal range} = R = \frac{v_0^2 \sin 2\alpha}{g}$$

The horizontal range is maximum if $\sin 2\alpha$ is maximum,

$$\sin 2\alpha = 1$$

$$2\alpha = \sin^{-1}(1)$$

$$2\alpha = 90^\circ$$

$$\alpha = 45^\circ$$

This shows that the horizontal range is maximum if the angle of projection is 45° . Put $\alpha = 45^\circ$ in formula to get

$$R_{\max} = \frac{v_0^2 \sin 2(45^\circ)}{g} = \frac{v_0^2 \sin 90^\circ}{g}$$

$$R_{\max} = \frac{v_0^2}{g}$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS



Subscribe

Q.No.3: Define Time of Flight?

ANSWER: The time taken by a projectile during its flight from point of projection to the point where it strikes the ground is called the **time of flight**.

Q.No.4: Define parabola of safety?

ANSWER: The locus of all points on different trajectories which lie at maximum distances from point of projection is called **parabola of safety**.

Q.No.5: Define central force?

ANSWER: If the $\vec{F}$ on a particle always passes through a fixed point O , the force is called a **central force** and O is called the **centre of force**.

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS

 [Subscribe](#)

Q.No.6: If 8 ft/s is the velocity of projection, then find the maximum horizontal range.

SOLUTION:

Velocity of projection = $v = 8 \text{ ft/s}$

By using formula of Maximum Horizontal range, we get

$$R_{\max} = \frac{v_0^2}{g}$$

$$R_{\max} = \frac{64}{32}$$

$$R_{\max} = 2 \text{ ft}$$

REQUIRED ANSWER



MTH404

SOLVED QUESTIONS



Q.No.7: A projectile is projected with speed 30 m/s at an angle of 30° to the horizontal, find the time of flight.

SOLUTION:

Speed of projectile = $v = 30 \text{ m/s}$

Angle of projectile = $\alpha = 30^\circ$

By using formula of Time of Flight, we get

$$t = \frac{2v_o \sin \alpha}{g}$$

$$t = \frac{2(30)(\sin 30^\circ)}{10}$$

$$t = \frac{60(0.5)}{10}$$

$$t = \frac{30}{10} = 3 \text{ s}$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS



Q.No.8: A particle describes the curve $a/r = n\theta$ under force f to the pole, show that the force is stated as $f \propto 1/r^3$.

SOLUTION:

$$\frac{a}{r} = n\theta$$

As we know that $u = \frac{1}{r}$,
so by putting value, we
get

$$au = n\theta$$

Taking derivative, we get

$$a \frac{du}{d\theta} = n$$

$$\frac{d^2u}{d\theta^2} = 0$$

Differential equation
of the orbit is

$$\frac{f}{h^2u^2} = \frac{d^2u}{d\theta^2} + u$$

$$\frac{f}{h^2u^2} = 0 + u$$

$$\frac{f}{h^2u^2} = u$$

$$f = h^2u^3$$

Here h^2 is constant
so by Putting the
value of u , we get

$$f = h^2 \left(\frac{1}{r} \right)^3$$

$$f \propto \frac{1}{r^3}$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS



Q.No.1: What is a projectile?

ANSWER: A **projectile** is any object that once projected or dropped continues in motion by its own mass and is influenced only by the downward force of gravity.

Q.No.2: Define horizontal range?

ANSWER: The maximum distance covered along horizontal direction by a projectile during its flight is called the **horizontal range**.

Q.No.3: Write the equation of orbits in polar coordinates.

ANSWER: Differential equation of the orbit in polar coordinates is

$$\frac{d^2u}{d\theta^2} + u = \frac{f}{h^2u^2}$$



MTH404

SOLVED QUESTIONS



Subscribe

Q.No.4: Define Apsidal distance?

ANSWER: An **apse** is a point on an orbit at which the radius vector drawn from the centre of force is a maximum or minimum. The length of the radius vector at such a point is known as the **apsidal distance**.

Q.No.5: A ball is thrown with a speed of **20m/s** and hits the horizontal ground **20m** away. Find the angle of projection.

SOLUTION:

Speed of ball = $v = 20\text{m/s}$

Range = $R_{\max} = 20\text{m}$

Angle of Projection = $\alpha = ?$

$$R = \frac{v_0^2 \sin 2\alpha}{g}$$

By Putting the values

$$20 = \frac{(20)^2 \sin 2\alpha}{10}$$

$$20 \times 10 = 400 \sin 2\alpha$$

$$\frac{200}{400} = \sin 2\alpha$$

$$\sin^{-1}(0.5) = 2\alpha$$

$$30^\circ = 2\alpha$$

$$\frac{30^\circ}{2} = \alpha$$

$$15^\circ = \alpha$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS

 Subscribe

Q.No.6: State and proof Kepler's Third Law.

Statement: The squares of the periods of the planets are proportional to the cubes of the semi-major axes of their orbits.

Proof: Since $\frac{1}{2}h$ is equal to the area described in a unit time, it follows that if T be the time that the particle takes to describe the whole ellipse, then

$$\frac{1}{2}hT = \text{area of the ellipse} = \pi ab$$

$$T = \frac{2\pi ab}{h} = \frac{2\pi ab}{\sqrt{h^2}} = \frac{2\pi ab}{\sqrt{l\mu}} \quad \therefore h^2 = l\mu$$

$$T = \frac{2\pi ab}{\sqrt{\mu} \sqrt{l}} = \frac{2\pi}{\sqrt{\mu}} \frac{ab}{\sqrt{b^2/a}} \quad \therefore l = \frac{b^2}{a}$$

$$T = \frac{2\pi}{\sqrt{\mu}} \frac{ab}{b/a^{1/2}} = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

$$T^2 \propto a^3$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS



Q.No.7: State Kepler's First Law.

ANSWER: Each planet describes an ellipse with the sun at one focus.

Q.No.8: State Kepler's Second Law.

ANSWER: The radius vector drawn from the sun to the planet sweeps out equal areas in equal times.

Q.No.9: Write the equation of orbits in Pedal Form.

ANSWER: Differential equation of the orbit in pedal form is

$$\frac{h^2}{p^3} \frac{dp}{dr} = f$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ



MTH404

SOLVED QUESTIONS

 Subscribe

Q.No.10: If R_1 and R_2 are the ranges of a projectile when it is projected at angles 30° and 75° respectively with the same speed of projection. Show that $R_1 = \sqrt{3}R_2$.

SOLUTION:

$$\theta_1 = 30^\circ$$

$$\theta_2 = 75^\circ$$

As we know that

$$\frac{R_1}{R_2} = \frac{\sin 2\theta_1}{\sin 2\theta_2}$$

$$\frac{R_1}{R_2} = \frac{\sin 2(30^\circ)}{\sin 2(75^\circ)}$$

$$\frac{R_1}{R_2} = \frac{\sin 60^\circ}{\sin 150^\circ}$$

As we know that
 $\sin \theta = \sin(180^\circ - \theta)$

$$\frac{R_1}{R_2} = \frac{\sqrt{3}/2}{\sin(180^\circ - 30^\circ)}$$

$$\frac{R_1}{R_2} = \frac{\sqrt{3}/2}{\sin 30^\circ}$$

$$\frac{R_1}{R_2} = \frac{\sqrt{3}/2}{1/2}$$

$$\frac{R_1}{R_2} = \frac{\sqrt{3}}{2} \times \frac{2}{1}$$

$$\frac{R_1}{R_2} = \sqrt{3}$$

$$R_1 = \sqrt{3}R_2$$

SOLVED BY: WAQAR HUSSAIN

SUBSCRIBE CHANNEL: TALEEMI MARKAZ